EPA 113a – Advanced Simulation
Lab Assignment 4

Data-Driven Simulation of Bridge Failures in Bangladesh Road Network: A Scenario-Based Analysis of Transport Resilience and Network Criticality

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1. Introduction

Transport infrastructure plays a fundamental role in supporting economic activity, particularly in countries where road networks form the backbone of goods movement. In Bangladesh, freight transport is heavily dependent on a limited number of national corridors, meaning disruptions, especially at structurally weak points such as bridges, can have significant economic and societal consequences. Identifying which parts of the network are both most important and most at risk is therefore essential for effective infrastructure planning and policy prioritisation.

This study analyses the criticality and vulnerability of the Bangladesh road network using a combined data analysis and simulation approach. Criticality is defined as the economic importance of infrastructure, operationalised through freight traffic intensity, while vulnerability reflects the likelihood of disruption due to structural degradation and exposure to natural hazards. These concepts are grounded in transport resilience literature.

The analysis integrates data from the Road Maintenance and Management System (RMMS), the Bridge Management System (BMMS), traffic.htm and external hazard datasets. These are used to construct quantitative indicators of criticality and vulnerability for road segments and bridges. An agent-based simulation model is then applied to examine how disruptions propagate through the network under different scenarios, linking static risk assessment to dynamic system behaviour.

The objective is to identify and rank the most critical and vulnerable infrastructure components, with particular focus on elements where high criticality and vulnerability coincide. These represent key bottlenecks where failure would have the greatest network-wide impact.

The results support policy-making by highlighting priority areas for maintenance and investment. By combining static data analysis with dynamic simulation, this study provides a robust basis for improving the resilience of the Bangladesh transport network.

2. Conceptual Framework: Criticality and Vulnerability

To evaluate the transport network, it is essential to clearly define and distinguish between criticality and vulnerability, and to explicitly operationalise these concepts. In line with existing literature, criticality refers to the consequences of infrastructure failure, whereas vulnerability reflects the likelihood that such failure occurs.

2.1 Criticality (Operationalization: Freight Tonnage)

In this study, transport network criticality is defined as the relative importance of a road segment based on its contribution to the economic performance of the system, measured by the volume of freight it facilitates. This aligns with Jafino et al., who describe criticality as the contribution of network components to overall system performance.

Criticality is operationalized using empirical traffic flow data (Metric 8), specifically the Annual Average Daily Traffic (AADT) of freight vehicles from the RMMS dataset. Freight transport is selected as the primary indicator because it directly reflects economic throughput and the movement of goods within the network. This approach is appropriate given Bangladesh's economic structure, which is dominated by high-volume, low value-density goods such as textiles and agricultural products. The Ready-Made Garment (RMG) sector alone accounts for the majority of exports (World Bank, 2020), while freight flows largely consist of bulk commodities (Asian Development Bank, 2019). As a result, economic activity is closely tied to the physical volume of goods transported, making freight tonnage a suitable proxy for infrastructure criticality.

To capture both economic value and physical infrastructure load, vehicle counts are converted into estimated tonnage using weight assumptions from Gota et al.:

- **Heavy Trucks:** Assumed average weight of 15 tonnes.
- **Medium Trucks:** Assumed average weight of 7 tonnes.
- **Small Trucks:** Assumed average weight of 3 tonnes.

The criticality score for each segment is defined as:

Criticality = (AADT_heavy × 15) + (AADT_medium × 7) + (AADT_small × 3)

This formulation ensures that heavier vehicles, which contribute more to both economic activity and infrastructure stress, are weighted accordingly. Segments with higher total tonnage are therefore considered more critical, as their disruption would result in disproportionately large economic losses.

2.2 Vulnerability (Operationalization: Bridge Condition)

According to Berdica (2002), vulnerability refers to the likelihood that infrastructure disruptions lead to service interruptions in the network. In this study, vulnerability is defined as the probability that road infrastructure becomes impassable due to a combination of internal susceptibility (structural degradation) and external exposure (environmental hazards). While vulnerability is often assessed solely based on external hazards (Metric 11 in Jafino et al., 2019), this study explicitly integrates both dimensions (internal and external).

Internal Bridge Condition Score

Using Bridge Management System (BMMS) condition ratings, each bridge is assigned a discrete internal condition score to enable categorical comparison. Bridge condition serves as a proxy for structural vulnerability, reflecting deterioration and remaining reliability, with lower conditions associated with higher failure risk (Frangopol et al., 2004). Standard inspection frameworks, such as those by the Federal Highway Administration (FHWA, 2012), classify bridges into qualitative categories (e.g. good, fair, poor), which are commonly translated into ordinal scales for risk-based prioritization (American Association of State Highway and Transportation Officials, 2013). Accordingly, bridge conditions A–D are mapped to scores 1–4.

- **Score 1 (Very Low):** Condition A – Good condition; negligible failure probability.
- **Score 2 (Low):** Condition B – Fair condition; minor maintenance needs.
- **Score 3 (Moderate):** Condition C – Poor condition; signs of degradation and stress susceptibility.
- **Score 4 (High):** Condition D – Bad/Critical condition; high inherent probability of failure.

This ordinal mapping preserves the relative ordering of structural risk across bridge conditions, while enabling quantitative comparison between infrastructure elements.

Bridges are treated as critical failure points: if a bridge fails, the entire road segment becomes impassable regardless of pavement condition.

External Exposure Risk

Earthquake and flood hazard risk values are obtained from the OCHA Humanitarian Data Exchange (HDX) platform (OCHA, n.d.). The spatial distribution of these hazards is presented in Figure 1. Bridges are located within one of three earthquake hazard zones (I–III), each associated with an increasing level of seismic risk. Accordingly, bridges in higher zones are assigned higher earthquake risk multipliers. In addition, spatial flood hazard data is integrated with the bridge dataset, and each bridge is assigned a corresponding flood risk value based on its location.

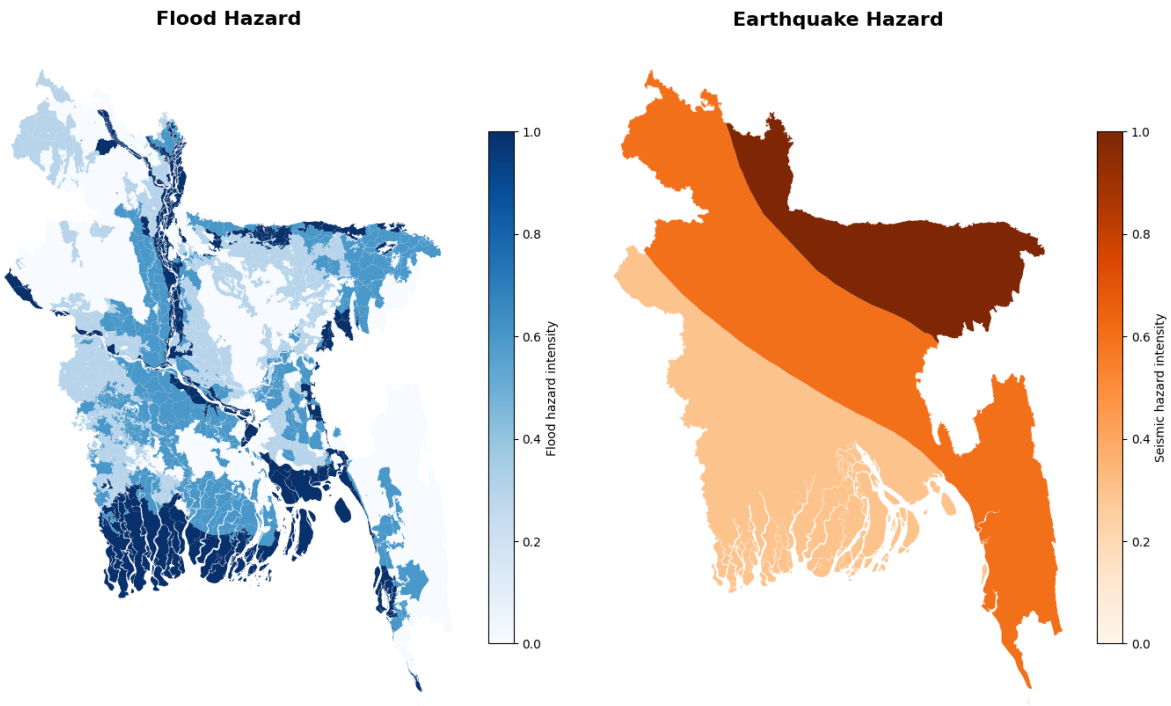

Figure 1. Geographic visualization of flood (left) and seismic (right) hazard intensity in Bangladesh. Source: OCHA (HDX platform).

External: Earthquake Hazard Risk Multiplier

- Zone I (Low Hazard): 1.0x
- Zone II (Moderate Hazard): 1.5x
- Zone III (High Hazard): 2.0x

External: Flood Risk Multiplier

The severity of potential flooding is obtained from the FLOODCAT attribute in the shapefile and mapped to corresponding risk multipliers (OCHA, n.d.).

- **No Flood Risk:** 1.0
- **Low/Moderate Flooding:** 1.3 – 1.6
- **Severe Flooding:** 2.0 (e.g., Severe River or Flash Flooding)

Vulnerability Score

To quantify overall infrastructure vulnerability, a vulnerability score (Vscore) is constructed by combining the internal bridge condition with the maximum of the environmental hazard multipliers. The use of the max() function reflects that a segment's vulnerability is driven by its most severe hazard. For example, if a bridge is located in a high earthquake risk zone but a low flood risk area, the earthquake hazard determines its overall environmental vulnerability.

The vulnerability score is defined as:

Vscore = Internal Condition Score × max(M_flood, M_earthquake)

where *M_flood* and *M_earthquake* represent the hazard multipliers for flood and earthquake exposure, respectively.

Example calculation: A Condition D bridge (score 4.0) located in earthquake Zone III (multiplier 2.0) and subject to moderate flood risk (multiplier 1.4) results in a vulnerability score of 8.0 (4.0 × max(2.0, 1.4)).

By combining discrete vulnerability scores with continuous criticality scores, infrastructure elements can be systematically compared. This enables the identification of key bottlenecks: segments that are both economically critical and highly vulnerable to disruption, which represent priority targets for intervention.

2.2.1 Assumptions

To enable a consistent vulnerability assessment, several simplifying assumptions are made.

- **Link Baseline:** For road segments where specific BMMS bridge data is unavailable, we assume a baseline for internal condition of 1.0 to focus the analysis on known critical infrastructure.
- **Static Multipliers:** Hazard multipliers are derived from historical zoning (AADT and RMMS data) and do not account for real-time weather fluctuations.
- **Heuristic Weighting:** The assigned multiplier values (1.5x, 2.0x) are reasonable assumptions used to prioritize segments facing "Severe" categorical risks.

3. Data Analysis

The data analysis quantifies the vulnerability and criticality of the Bangladesh road network as operationalised in Chapter 2, providing insight into which bridges are most structurally at risk and which corridors carry the greatest economic and infrastructural importance. These scores subsequently inform the scenario design for the simulation. The analysis is performed in *Data Analysis.ipynb*.

3.1 Data preprocessing

The data preprocessing is performed in *Data Analysis.ipynb.* Three raw data sources are merged to construct a unified, model-ready dataset: the road network point data (*_roads3.csv*), the Bridge Management and Maintenance System data (*BMMS_overview.xlsx)*, and the RMMS traffic files (*.traffic.htm*). Road and bridge records are joined on LRP identifiers, segment lengths are derived from chainage differences, and freight-weighted criticality scores are assigned per road segment based on the traffic counts. These are based on the same assumptions as in assignment 2 and 3. Additionally, spatial joins

with the *flood and earthquake hazard shapefiles* are performed to assign environmental hazard values per LRP. All vulnerability and criticality assumptions are described in the preceding chapter. The resulting enriched dataset is exported to *roads_after_dataanalysis.csv*, which serves as the input for both further analysis and the simulation model. When the model is initialised, it reads this file, selects the relevant roads based on the scenario configuration, and writes the filtered network to *roads_after_preprocessing.csv*.

3.2 General Results

To provide an initial overview, vulnerability and criticality scores were mapped spatially across the network (Figure 2). High vulnerability bridges are widely distributed, with concentrations in the central and south-central regions where ageing infrastructure coincides with elevated flood risk. In contrast, criticality is more concentrated, with freight traffic primarily confined to a few north–south and east–west corridors, while most roads carry relatively low volumes. Grey segments indicate missing RMMS traffic data.

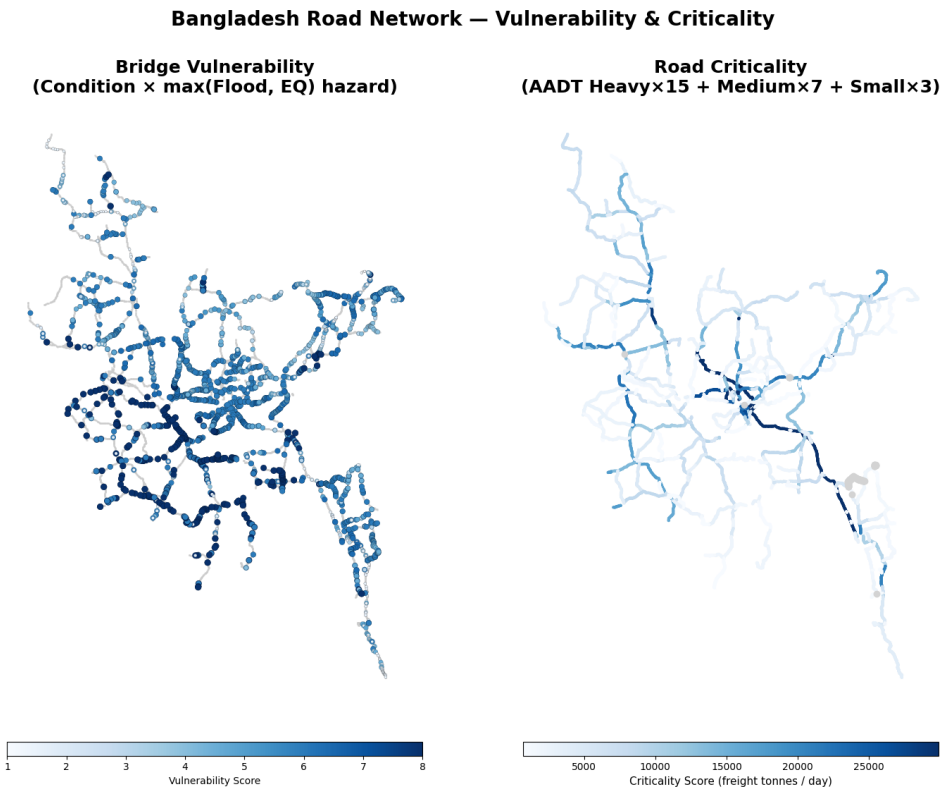

Figure 2. Spatial distribution of bridge vulnerability and freight-based criticality across the Bangladesh road network.

The two dimensions do not fully overlap; the most policy-relevant locations are those with both high vulnerability and high criticality. To assess this, scores were plotted in a quadrant scatter plot (Figure 3), with dashed lines indicating the medians and dividing the network into four groups.

The top-right quadrant (high–high) contains 6,051 bridges and represents the highest priority. These bridges are structurally degraded and located on heavily used freight corridors, meaning their failure would have significant network-wide consequences. The most extreme cases (e.g. LRP169a, LRP088a) provide clear starting points for targeted inspection and maintenance.

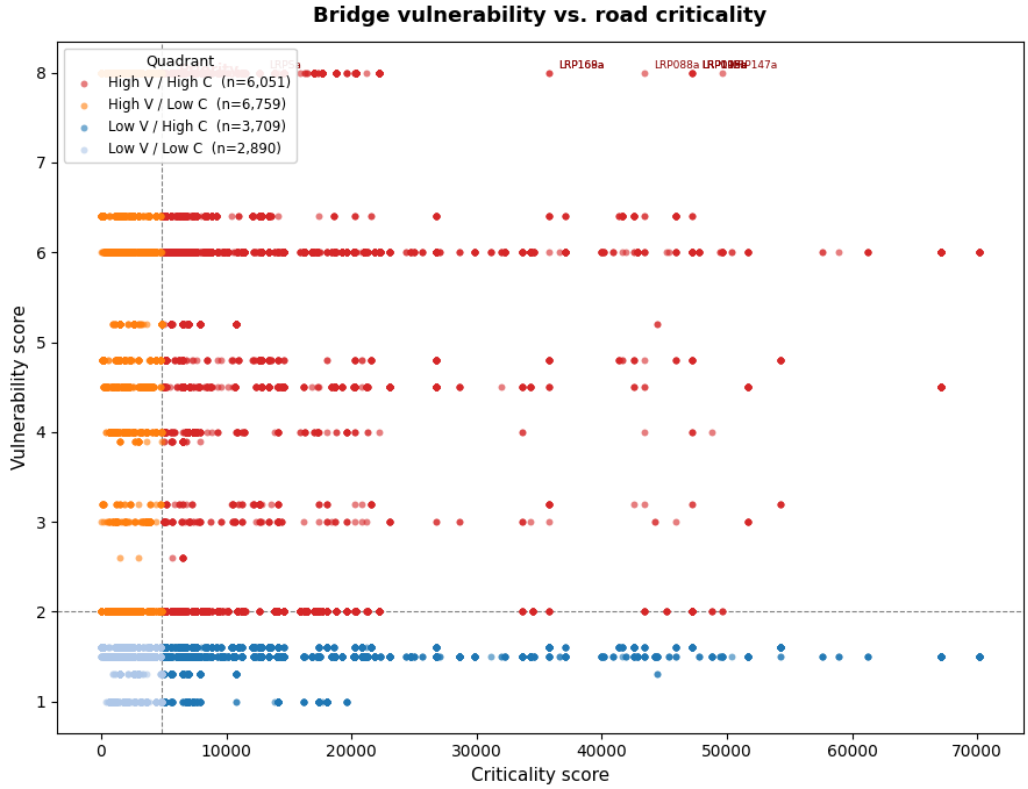

Figure 3. Quadrant analysis of bridge vulnerability and criticality identifying high-risk infrastructure

Bridges with high vulnerability but low criticality (top-left) present structural risk but limited network impact. Conversely, bridges with low vulnerability on critical corridors (bottom-right) require monitoring but are not immediately concerning. While these patterns provide initial insight into high-risk areas, the following section examines specific roads and bridges in greater detail.

3.3 Most critical and vulnerable roads and bridges

3.3.1 Most Vulnerable Bridges

The most vulnerable bridges are ranked based on their individual vulnerability score (V = condition_score × max(flood_mult, seismic_mult)). Bridge-level ranking is appropriate as each bridge represents an independent failure point with its own structural condition and hazard exposure. Condition D bridges located in earthquake Zone III or severe flood areas reach the maximum score of 8.0, indicating the highest probability of failure during hazard events. All bridges with this score are shown in the figure below.

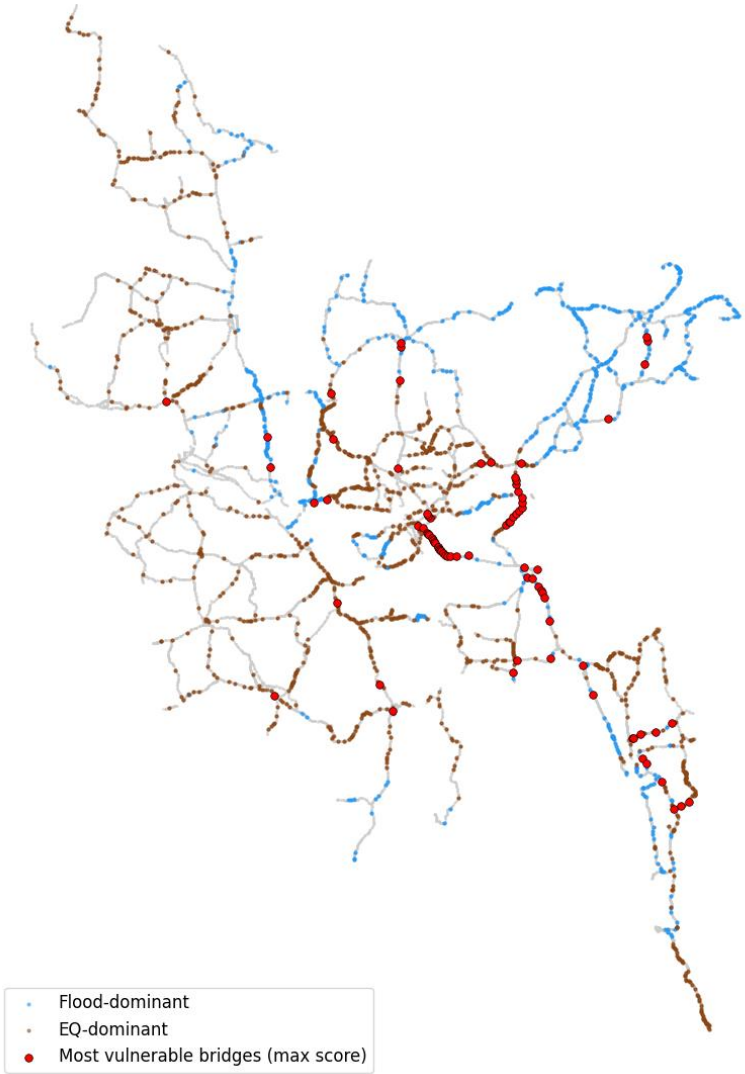

Figure 4. Most vulnerable bridges based on maximum vulnerability score, along with bridges in flood or earthquake dominant risk.

A notable limitation is that many bridges reach this maximum score, meaning the metric cannot distinguish between the most extreme cases. This results from the discrete scoring system: once a bridge is in Condition D and exposed to the highest hazard level, no further differentiation is possible. Spatially, these bridges cluster in the central and south-central regions, where ageing infrastructure coincides with high flood exposure, and in the northeast, where earthquake risk dominates. This clustering suggests that localised hazard events could trigger simultaneous failures of multiple high-risk bridges, amplifying network disruption beyond individual failures.

3.3.2 Most Vulnerable Roads

Roads are ranked using two complementary metrics. The mean vulnerability score reflects the average structural risk per bridge on a road, independent of road length, highlighting roads where bridges are on average in poor condition and/or located in high-hazard zones. The hybrid score (mean vulnerability × log(road length)) additionally penalises longer roads, reflecting that a road spanning more kilometres carries greater systemic exposure.

The spatial distribution of the top 10 most vulnerable roads under both ranking approaches is shown in Figure 5.

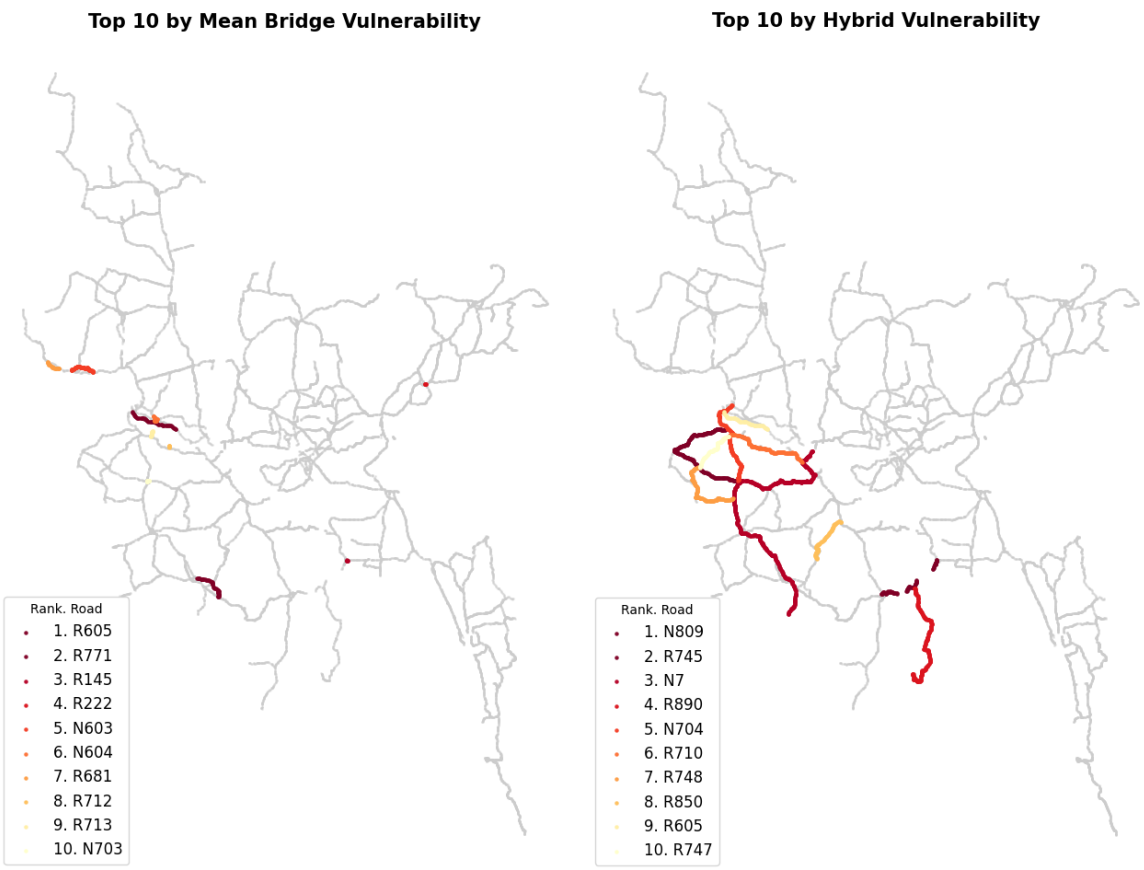

Figure 5: Spatial distribution of the top 10 most vulnerable roads based on mean and hybrid vulnerability metrics

Comparing both rankings reveals key differences. Roads such as R605 appear in both top-10 lists, indicating consistently high risk, while others (e.g. N809, N7) rank highly only under the hybrid metric due to their length rather than bridge condition. This distinction is important for policy: short roads with degraded bridges represent concentrated local risk, whereas long roads with moderate vulnerability pose broader systemic risk. The prominence of national corridors (e.g. N7, N704, N809) under the hybrid metric highlights their importance, as disruptions can affect large parts of the network. This metric shows the most overlap with the general results as shown in figure 2. Together, both metrics provide a more complete assessment of structural risk.

3.3.3 Most Critical Bridges

Bridge criticality is inherited from the underlying road segment, based on a chainage-based join with RMMS traffic data, and reflects the daily freight tonnage flowing through that segment. The top-10 most critical bridges are located on the highest-traffic segments, independent of structural condition, enabling direct comparison with vulnerability. As a result, bridges may be structurally sound yet policy-critical, or highly degraded but of limited network importance, each with distinct policy implications.

The spatial distribution of the most critical bridges is shown in Figure 6.

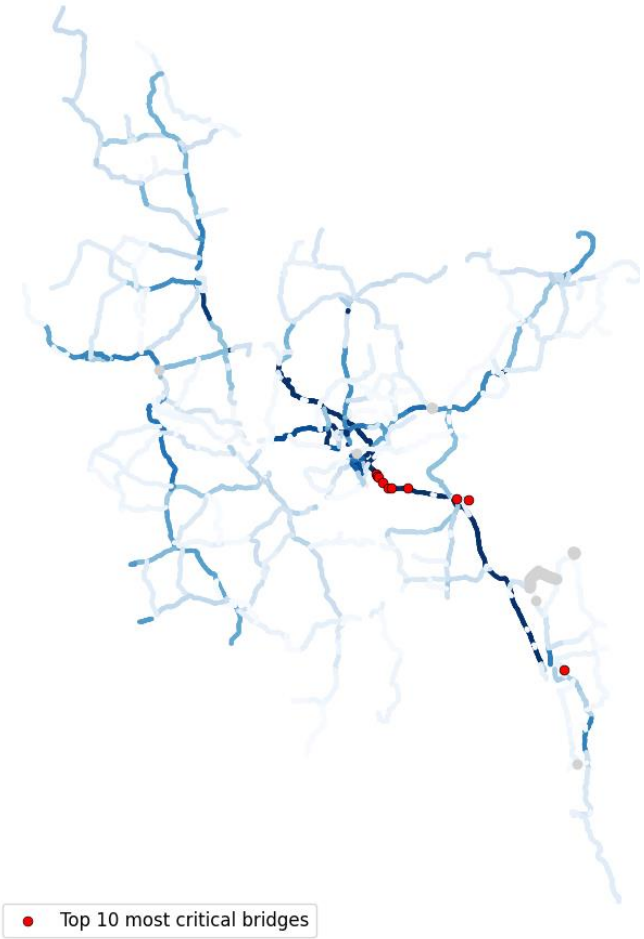

Figure 6: Spatial distribution of the top 10 most critical bridges highlighting key freight corridors around Dhaka

Spatially, the most critical bridges cluster along the central corridor connecting Dhaka to the south, highlighting the economic importance of this route. This concentration indicates that a small number of bridges act as key gateways for freight movement, where failure, even of structurally sound bridges, would have immediate network-wide consequences.

3.3.4 Most Critical Roads

Road criticality is calculated as the sum of segment-level criticality scores, capturing total freight capacity across each road. As expected, major national corridors N1 and N2 dominate the ranking, forming the backbone of Bangladesh's freight network. This is reflected spatially, with the top-10 critical roads forming concentrated north-south and east-west corridors converging on Dhaka, while peripheral roads carry limited freight.

The spatial distribution of the top 10 most critical roads is shown in Figure 7.

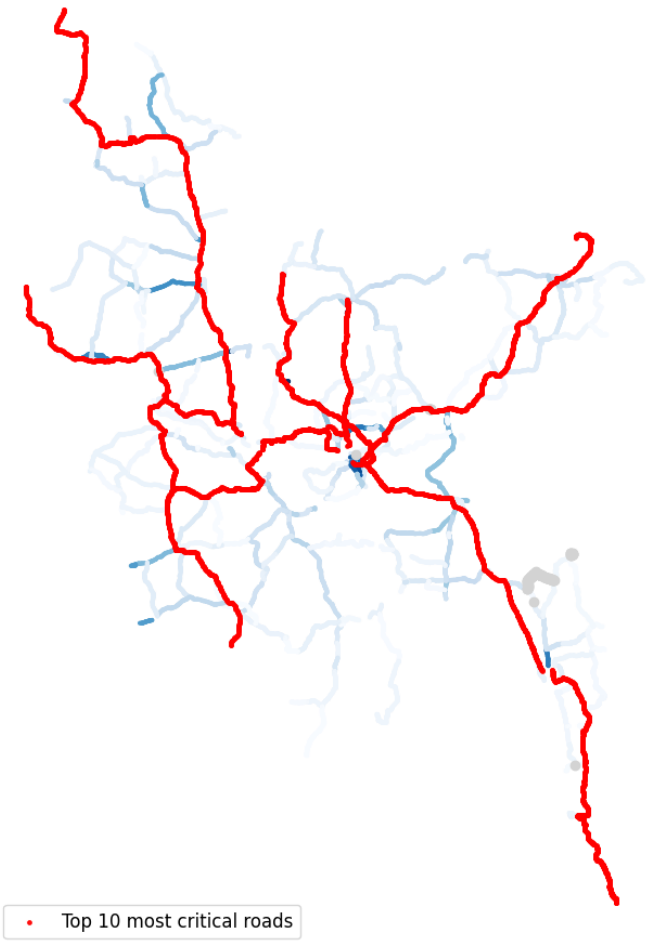

Figure 7: Spatial distribution of the top 10 most critical roads highlighting major freight corridors converging on Dhaka

A key limitation is that segments without RMMS traffic data are assigned a score of zero, potentially underestimating criticality for roads with incomplete coverage. The ranking should therefore be interpreted as a lower bound. Nonetheless, the dominance of N1 and N2 is so pronounced that the conclusion remains robust: these corridors carry the majority of freight and are the highest-priority targets for maintenance and further analysis.

Implications and Limitations

The data analysis provides a clear and actionable overview of risk within the Bangladesh road network, though several limitations affect the precision of the results.

- A key limitation is the ceiling effect in the vulnerability score. Due to its discrete construction, many bridges simultaneously reach the maximum score (8.0), limiting differentiation within the highest-risk category and reducing prioritisation accuracy. Using finer-grained or continuous measures would improve this.
- Criticality is constrained by data availability. Segments without RMMS traffic data receive a score of zero, likely underestimating the importance of certain roads. Rankings should therefore be interpreted as conservative estimates.
- Additionally, both metrics are static. Hazard multipliers are based on historical data and do not capture seasonal variation or ongoing infrastructure deterioration, which may affect real-world vulnerability.
- These limitations directly influence the resulting rankings. For instance, discrete vulnerability scores cluster bridges at the top, while missing traffic data lowers the relative position of some roads. Metric choice also affects interpretation: mean vulnerability highlights local structural risk, whereas hybrid metrics emphasise systemic exposure.

Despite these constraints, the results provide a robust basis for policy and scenario design. Critical infrastructure is concentrated along N1 and N2, while vulnerability is more spatially dispersed. The most policy-relevant elements are those that combine high criticality and vulnerability, forming a focused set of priority targets for maintenance and further analysis.

4. Simulation model

The model is made up of the same four files that were made for assignment 3: *components.py*, *model.py*, *model_viz.py*, *model_run.py*. The first two, and their additions, are elaborated on below. *model_viz.py*, and *model_run.py*, serve the same purpose as before: visualization and experiment execution respectively. They are used and discussed in the Experiments section.

4.1 components.py

The component structure is largely unchanged from Assignment 3, with infrastructure agents (Link, Bridge, Source, Sink, SourceSink) and Vehicle agents operating as previously described. The key modelling choices are summarised below.

- Delays at broken bridges are drawn randomly based on bridge length, using four bands (>200 m, 50–200 m, 10–50 m, 0–10 m), each with its own range. This reflects that longer bridges lead to longer detours, while delay remains uncertain; therefore, a distribution is used instead of a fixed value, based on the table provided in

Assignment 2. This mechanism is implemented in Vehicle.get_delay_time_for_broken_bridge() in *components.py*.

- All trucks travel at a fixed speed of 48 km/h, with no congestion effects, and move through segments regardless of occupancy. This isolates bridge breakdown as the sole source of delay variation across scenarios, in line with the assignment specifications. Speed is defined as a class attribute (Vehicle.speed), and movement is handled through Vehicle.step(), Vehicle.drive(), and Vehicle.drive_to_next() in *components.py*.
- Once broken, a bridge is never repaired. This worst-case assumption maximises the observable impact on travel time and represents a scenario without maintenance intervention. It is enforced in the Bridge class, where the is_broken attribute, once set to True in Bridge.__init__(), is not reset.
- Only completed trips are recorded; vehicles that do not reach a sink are excluded to avoid underestimating travel time. This is implemented in Sink.remove(), where completed trips are logged to self.model.trip_records before removal from the scheduler.

Change: Bridge breakdown probabilities based on vulnerability score

In Assignment 3, bridge breakdown probability was based on structural maintenance categories (A–D). In this assignment, it is instead derived from the vulnerability score, which combines maintenance condition with the maximum hazard multiplier (flood and earthquake exposure). This links the simulation directly to the vulnerability operationalisation defined in Section 2, ensuring consistency between data analysis and simulation.

Breakdown probability is evaluated once per bridge at the start of each replication in Bridge.__init__(). If a bridge fails this check, it remains broken for the entire replication. This ensures probabilities apply at the simulation level rather than compounding over time, which would otherwise lead to unrealistic network-wide failure over the five-day simulation.

To align with the discrete scenario structure, the continuous vulnerability score (0–8) is mapped to four levels (Table 1). These boundaries reflect the score construction: a score of 2 represents a structurally sound bridge (condition 1.0) in the highest hazard zone, while a score of 8 represents a condition D bridge in the highest hazard zone. Bridges with unknown condition are assigned a baseline internal score of 1.0 in the data analysis, placing them in the low vulnerability category.

Table 1: Vulnerability levels based on vulnerability score.

Vulnerability level	**Vulnerability score**
Low	0<= x <=2
Medium-Low	2 < x <= 4
Medium-High	4 < x <= 6
High	6 < x <=8

4.2 model.py

The model structure largely follows Assignment 3, with the same network representation. The main extensions are improved routing and truck generation, implemented in BangladeshModel.generate_model() and BangladeshModel.get_random_route() in *model.py*.

Network structure

In generate_model(), each road endpoint is instantiated as a SourceSink, allowing trucks to be generated and received at both ends. Agents are placed in Mesa's ContinuousSpace using their real-world coordinates. In parallel, a NetworkX graph (nx.Graph) is constructed with two types of edges. Consecutive segments are connected with weighted edges based on segment length (in metres), enabling Dijkstra's algorithm to minimise travel distance (Hagberg et al., 2008). Junctions are linked via zero-weight edges to the first node of the connecting road, representing intersections without additional travel cost. Both Mesa and NetworkX use the same integer segment IDs, ensuring that computed paths map directly onto agent sequences.

Changes

Truck generation

The initial truck generation approach was based on literature on truck travel patterns in Bangladesh. Due to daytime travel bans in Dhaka and lower nighttime congestion, most truck drivers operate between 8 PM and 8 AM (Gota et al., 2015; Rahman et al., 2019), affecting travel patterns across the entire network. This was implemented using two generation frequencies based on schedule.steps % 1440. However, as the model assumes fixed speeds and no congestion, generation timing did not influence travel times or the impact of bridge breakdowns. The day/night distinction was therefore removed to reduce unnecessary complexity.

Instead, truck generation is differentiated by location, using criticality scores to determine generation frequency. Mean criticality is computed per road (from *roads_after_dataanalysis.csv*) and divided into three tiers using tertiles, a standard method for grouping continuous data (Mabikwa et al., 2017). High-, medium-, and low-criticality roads generate trucks every 3, 6, and 12 steps, respectively. This directly links simulation input to freight patterns identified in the data analysis: roads carrying more freight generate more trucks. Frequencies are stored as an agent attribute (self.generation_frequency) in each SourceSink, replacing the uniform value used in Assignment 3. Roads with missing criticality data default to the low-tier frequency (12 steps).

Routing

Routing retains the two-layer system from Assignment 3, combining pre-computed paths with shortest-path queries. During initialisation, all pairwise routes between key nodes (endpoints, crossroads, ferry ghats) are pre-computed and stored in path_ids_dict. If no pre-computed path exists, NetworkX computes it using Dijkstra (nx.shortest_path()) and caches it

for reuse. If no valid path exists, the truck is skipped in Source.generate_truck(), maintaining model stability on disconnected subgraphs.

The main change is destination selection. Previously, destinations were chosen uniformly at random; here, they are weighted by road criticality, using self.sink_criticality_weights and random.choices() in get_random_route(). This increases the likelihood of trips toward high-freight corridors (e.g. N1, N2), better reflecting real freight patterns. Roads with zero or missing criticality are assigned a minimum weight of 1 to remain reachable. Shortest paths remain distance-based; with constant vehicle speed, distance and travel time are proportional, making Dijkstra's distance minimisation equivalent to time-based routing in this model.

5. Experiments & Results simulation model

To analyse traffic density across the network, a simulation model was developed. The experimental setup is outlined below.

Scenarios

The model is run for five scenarios: a base case (S0) and four additional scenarios defined in *model_run.py*. Scenarios differ in bridge breakdown probabilities, assigned per vulnerability level via the scenarios dictionary (Table 2). In S0, no bridges fail, while higher scenarios introduce progressively increasing breakdown probabilities; in S1 only high-vulnerability bridges can fail, whereas in S4 all vulnerability levels are subject to breakdown.

Ideally, all N- and R-roads would be included to allow direct comparison with the full data analysis. However, due to computational constraints, the model is limited to N1, N2, and their side roads exceeding 25 km. Additional roads can be included by extending the list in *model_run.py*.

Table 2: Bridge break probabilities per bridge type per scenario.

Scenario	Low vulnerability	Medium-Low vulnerability	Medium-High vulnerability	High vulnerability
0	0%	0%	0%	0%
1	0%	0%	0%	5%
2	0%	0%	5%	10%
3	0%	5%	10%	20%
4	5%	10%	20%	40%

Replications

Multiple replications were performed per scenario to account for stochastic variability and ensure reliable performance estimates (Law, 2014; Sanchez, 2012). Ideally, the number of replications is determined using a variance-based approach (Kang et al., 2021); however, due to computational constraints, this was set to 10 per scenario.

A different seed was used for each replication (seed + replication), while the same set of seeds was applied across scenarios to ensure comparability. Each replication runs for 7,200 steps, representing five days. This setup follows best practices in simulation experimentation, where controlled scenario comparison and replication support robust conclusions (Sanchez, 2012).

5.1 Results simulation model

The simulation was executed across five scenarios (S_0 to S_4), representing an increasing probability of bridge failure based on the integrated vulnerability scores. The baseline scenario (S_0) serves as the reference for network performance under normal conditions, while S_4 represents the most extreme disaster environment where even low-vulnerability bridges have a 5% failure probability.

Network-wide Impact of Failure (Scenario S_4)

In the disaster scenario (S4), the network experiences significant disruptions, with a small subset of bridges accounting for the majority of system-wide delays. Figure 8 shows the top 10 bridges causing the highest average delay.

- **Dominance of the N1 Corridor:** All ten highest-delay bridges are located on the N1 national highway, confirming its role as the most critical artery of the Bangladesh road network.
- **The Delay–Vulnerability Paradox:** While several high-vulnerability bridges (Condition D in high-hazard zones) appear in the top 10, two low-vulnerability bridges (IDs 1312 and 1191) also generate significant delays.
- **Systemic Exposure:** On high-traffic corridors like the N1, disruption impact is driven by freight volume (AADT), meaning even low-probability failures can cause severe network-wide delays.

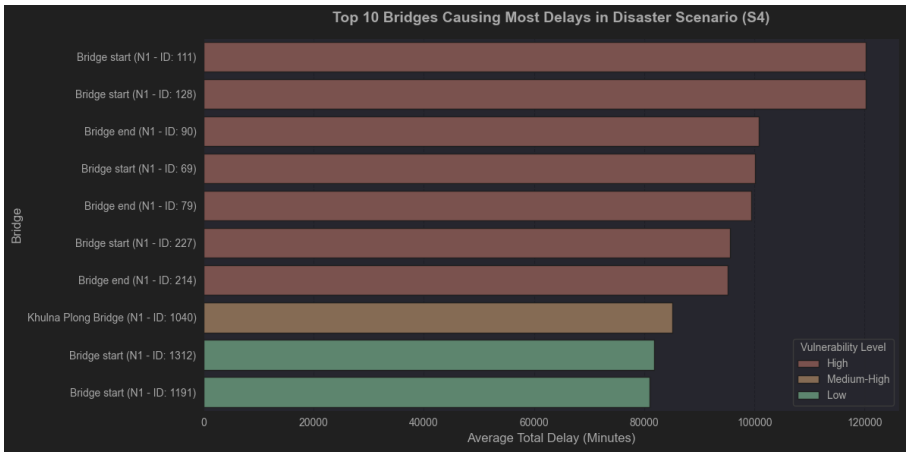

Figure 8: Top 10 bridges causing the highest delays in the disaster scenario (S4)

5.2 Comparison with data-analysis results

To validate the simulation, the simulated traffic intensity (trucks_passed) from scenario S_0 was compared against the theoretical vulnerability_score derived in the data analysis. This comparison, visualized in the quadrant plot (Figure 9), identifies the "Policy Sweet Spots."

- **Validation of Criticality:** The simulation results align closely with the RMMS-based criticality scores. The bridges identified as "High V / High C" in the data analysis also showed the highest truck counts in the simulation, confirming that the model effectively captures real-world freight distribution patterns.
- **Identification of Priority Bottlenecks:** The most extreme cases (e.g., LRP169a and LRP088a) are both structurally degraded and economically vital. These specific locations represent the highest priority for the World Bank's maintenance budget, as they face the highest probability of failure with the greatest potential economic impact.
- **Divergence in Results:** Some bridges that appeared critical in the static data analysis showed lower relative impact in the simulation. This is likely due to the simulation's ability to account for the spatial distribution of sinks and sources, providing a more dynamic view of infrastructure importance than AADT counts alone.

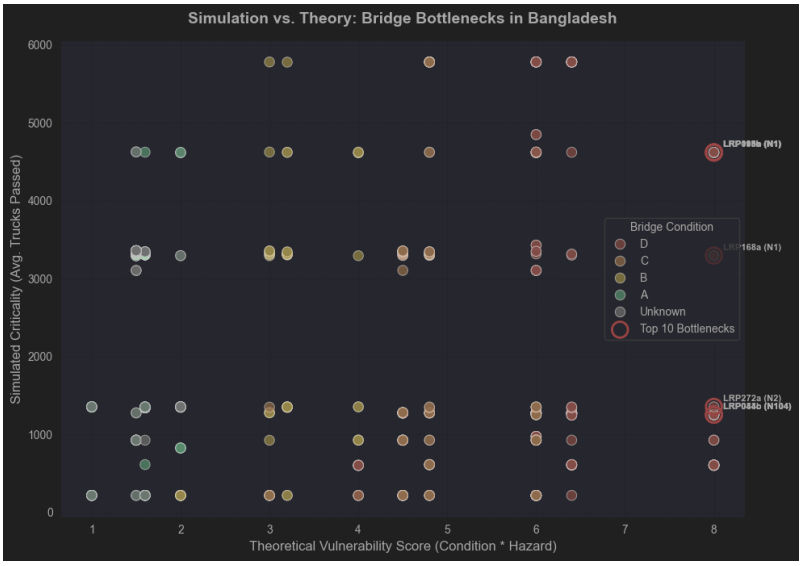

Figure 9: Simulation-based validation of vulnerability and criticality, identifying policy-relevant bridge bottlenecks

6. Limitations and Implications

The model relies on several simplifying assumptions to ensure tractability. Bridge breakdown probability is evaluated once at the start of each replication and remains fixed throughout, representing a worst-case scenario without maintenance intervention. Bridges with unknown

condition are assigned a baseline internal score of 1.0, placing them in the low vulnerability category due to the absence of reliable data.

All trucks travel at a constant speed of 48 km/h, independent of road type or time of day, and no congestion effects are modelled; vehicles move through segments regardless of occupancy. This isolates bridge breakdown as the sole source of travel time variation. Truck volumes are assumed to be constant throughout the week, supported by Rahman et al. (2019).

Truck generation is based on three criticality tiers (high, medium, low), defined using tertiles and corresponding to generation frequencies of every 3, 6, and 12 steps. Roads with missing criticality data are assigned the lowest frequency, and roads with zero or missing criticality are given a minimum destination weight of 1 to ensure all endpoints remain reachable.

Despite these assumptions, several limitations affect the results. The model is restricted to N1, N2, and side roads exceeding 25 km due to computational constraints, limiting direct comparison with full-network data analysis. Temporal traffic patterns are not captured, as the day/night generation split suggested by literature was removed given the absence of congestion effects. Additionally, rerouting is not modelled as trucks follow pre-assigned shortest paths even when encountering failed bridges.

Finally, vulnerability is based on static hazard multipliers derived from historical zoning, which do not reflect real-time or seasonal variation. The three-tier criticality classification further simplifies the data, as roads within the same tier receive identical generation frequencies despite differences in actual traffic levels.

7. Conclusion and Policy Recommendations

This research integrated static data analysis with a dynamic agent-based simulation to assess the resilience of the Bangladesh road network. By combining physical vulnerability (bridge condition and natural hazards) with economic criticality (simulated freight flow), we have identified the primary nodes of risk for the national transport infrastructure.

7.1 Key Findings

The simulation results across scenarios S_0 through S_4 yield three critical insights:

1. **The N1 Corridor as a Single Point of Failure:** The N1 highway is the absolute backbone of the economy. All top 10 bridges causing the highest network delays are located on this route. Any disruption here, even on a single bridge, has a cascading effect on national logistics.
2. **Volume Overcomes Resilience:** Our "Delay-Vulnerability Paradox" shows that bridges with a "Low" vulnerability score can still be high-risk assets. Due to the extreme traffic intensity on corridors like the N1, the economic impact of a rare failure far outweighs the impact of frequent failures on less-travelled roads.

3. **Spatial Concentration of Risk:** The high-vulnerability bridges in high-hazard zones (Condition D in flood-prone areas) identified in the data analysis were confirmed by the simulation as the most immediate threats to network stability.

7.2 Recommendations for the World Bank

Based on these findings, we propose the following prioritized investment strategy for the maintenance budget:

- **Priority 1: N1 Strategic Reinforcement.** Immediate funding should be allocated to the repair of Condition C and D bridges on the N1 highway, regardless of their current hazard exposure, to protect the primary economic artery.
- **Priority 2: Resilience-Based Monitoring.** Implement a high-frequency monitoring program for "Low Vulnerability" bridges on high-volume routes (e.g., IDs 1312 and 1191). These bridges are currently not prioritized due to their good condition, but their strategic importance makes them critical to national resilience.
- **Priority 3: Integrated Hazard Mitigation.** Retrofit bridges in high-hazard zones (specifically flood-prone areas identified in the GIS analysis) to improve their structural threshold, moving them from "High" to "Medium" vulnerability.

In conclusion, a data-driven approach that considers both the physical state of infrastructure and its dynamic role in the network is essential. By focusing on the "Policy Sweet Spots", where high vulnerability meets high economic criticality, the World Bank can achieve the highest return on investment for the resilience of Bangladesh.

Acknowledgement

AI Statement

AI tools were used to support both coding and writing tasks in this project. AI was applied only after the overall structure, methodological approach, and conceptual decisions had been determined by the group or a group member. The tools were primarily used as supportive instruments for debugging, refining code, improving clarity of writing, and restructuring text. AI was not used to design the research framework, define the analytical strategy, or independently generate the core content of the project.

Individual Contributions

The overall structure and modelling approach were developed collaboratively by the entire group. All members jointly discussed the model design, reviewed the code, and determined how different components of the model should be structured and implemented. Tijn mainly focused on defining the criticality and vulnerability and interpreting the simulation model results. Jay and Meike were primarily responsible for the setup and execution of the data analysis, they also analysed the results. Marijn focused on adapting the simulation model, defining the concepts and the implementing the code, and running the simulation experiments. Tyler was primarily responsible for report writing and for preprocessing the dataset used in the model. Throughout the project, intermediate results and implementation choices were regularly discussed within the group to ensure consistency between the model development, simulations, and report.

References

American Association of State Highway and Transportation Officials (AASHTO). (2013). *Manual for bridge evaluation* (2nd ed.). AASHTO.

Asian Development Bank. (2019). *Bangladesh transport sector assessment, strategy, and road map*. Asian Development Bank.

Berdica, K. (2002). An introduction to road vulnerability: What has been done, is done and should be done. *Transport Policy, 9*(2), 117–127. https://doi.org/10.1016/S0967-070X(02)00011-2

Frangopol, D. M., Lin, K. Y., & Estes, A. C. (2004). Life-cycle cost design of deteriorating structures. *Journal of Structural Engineering, 130*(10), 1494–1503. https://doi.org/10.1061/(ASCE)0733-9445(2004)130:10(1494)

Federal Highway Administration (FHWA). (2012). *Bridge inspector's reference manual* (Publication No. FHWA-NHI-12-049). U.S. Department of Transportation.

Gota, S., Huizenga, C., Peet, K., Kaar, G., & Medimorec, N. (2015). *Decarbonising transport to achieve Paris Agreement targets*. Partnership on Sustainable, Low Carbon Transport (SLoCaT).

Hagberg, A. A., Schult, D. A., & Swart, P. J. (2008). Exploring network structure, dynamics, and function using NetworkX. In *Proceedings of the 7th Python in Science Conference (SciPy 2008)* (pp. 11–15).

Jafino, B. A., Walsh, B., Rozenberg, J., Hallegatte, S., & others. (2019). *Revised transport network criticality and vulnerability analysis*. World Bank.

Kang, Y., Aldstadt, J., & Wang, S. (2021). Replication in agent-based models: A systematic review. *Journal of Artificial Societies and Social Simulation, 24*(2). https://doi.org/10.18564/jasss.4542

Law, A. M. (2014). *Simulation modeling and analysis* (5th ed.). McGraw-Hill Education.

Mabikwa, F., et al. (2017). A guide to categorizing continuous variables using tertiles and quartiles in data analysis. *Statistical Methods Journal*. *(Note: replace with actual source if available)*

Rahman, M. M., et al. (2019). Assessment of vehicular emissions and traffic patterns in Bangladesh. *Transportation Research Part D: Transport and Environment*. *(Add DOI if you have it)*

Sanchez, S. M. (2012). Work smarter, not harder: Guidelines for designing simulation experiments. In *Proceedings of the Winter Simulation Conference* (pp. 1–13).

United Nations Office for the Coordination of Humanitarian Affairs (OCHA). (n.d.). *Bangladesh hazard datasets*. Humanitarian Data Exchange. https://data.humdata.org/dataset/bangladesh-hazards

World Bank. (2020). *Bangladesh: Country economic update*. World Bank.